Project Proposal: Reproducing OWL and Testing Whether Outlier Weighed Layerwise Sparsity Still Helps Small Modern LLMs

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1 Introduction

We will reproduce "Outlier Weighed Layerwise Sparsity (OWL): A Missing Secret Sauce for Pruning LLMs to High Sparsity" (Yin et al., 2024). The paper studies one-shot unstructured pruning of LLMs with no retraining. Strong methods such as SparseGPT (Frantar and Alistarh, 2023) and Wanda (Sun et al., 2024) prune every layer to the same sparsity. OWL questions this choice and links pruning quality to outlier features, a few hidden dimensions with very large activations (Kovaleva et al., 2021; Dettmers et al., 2022).

OWL scores each weight with the Wanda metric $A_{ij} = \|\mathbf{X}_j\|_2 \cdot |W_{ij}|$. For each Transformer block l it computes the Layerwise Outlier Distribution (LOD) D^l, the fraction of weights whose score is over M times the block mean. Blocks with more outliers are pruned less: $S_l \propto 1 - D^l$, rescaled into a band of width 2λ whose mean is the target S.

We target four claims. (C1) OWL improves Wanda and SparseGPT at high sparsity. At 70% on LLaMA-7B, WikiText perplexity drops from 85.77 to 24.55 with Wanda and from 26.30 to 19.49 with SparseGPT, although OWL hurts SparseGPT on OPT-6.7B (20.29 to 22.48). (C2) With Wanda, the gain grows as models shrink: 61.22 points at 7B but only 6.62 at 65B. (C3) Zero-shot accuracy also improves on seven tasks. (C4) The LOD of dense LLaMA models is far from uniform and roughly U-shaped, and good pruning methods keep outliers.

2 Reproducing Main Results

We reproduce C1, C3, C4 and the 7B and 13B part of C2. We skip the 30B and 65B models, the CPU speedup study, LoRA fine-tuning and the appendix studies.

Models. Our main model is LLaMA-7B (Touvron et al., 2023). The README's checkpoint was removed from Hugging Face, so we use `huggyllama/llama-7b`. We also run OPT-6.7B (Zhang et al., 2022) and LLaMA-13B, with the paper's hyperparameters (M=5, λ=0.08 for 7B, M=7 for 13B, M=10 for OPT).

Data. Calibration uses 128 random 2048-token sequences from the first C4 shard (Raffel et al., 2020). Perplexity uses 2048-token windows of the WikiText-2 (Merity et al., 2017) test split, which the paper calls validation. Zero-shot accuracy covers BoolQ (Clark et al., 2019), RTE (Wang et al., 2018), HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2021), ARC-e, ARC-c (Clark et al., 2018) and OpenBookQA (Mihaylov et al., 2018). For R3 we run the authors' own zero-shot scripts (lm-eval-harness 0.3.0, Gao et al., 2024) on the saved pruned model. Table 4 does not say whether it reports acc or acc_norm, so we log both.

Experiments. (R1) Table 3: magnitude pruning, plus Wanda and SparseGPT each with uniform and OWL sparsity, at 70% on all three models, including the OPT-6.7B case where OWL hurts. (R2) Figure 2 and Table 8: a 40% to 80% sweep on LLaMA-7B with Wanda, comparing OWL with uniform, global, ER, ER-plus and OWL-inverse allocations. (R3) Table 4: zero-shot accuracy of the dense and 70% sparse LLaMA-7B models. (R4) Table 1 and Figure 1: the LOD of each dense model, recomputed after pruning with the dense model's threshold.

Success criterion. We first check the dense LLaMA-7B perplexity of 5.68 and report our own baselines next to the paper's. A claim counts as reproduced if every R1 setting shows the paper's direction of OWL gain or loss and our 70% perplexities are within 15% of the paper's. Methods within 2% count as tied.

3 Plans for Extension

C2 was only tested on LLaMA-1 (7B to 65B), trained on at most 1.4T tokens. Modern small models see far more data, about 15T tokens for Llama 3 (Grattafiori et al., 2024) and 18T for Qwen2.5 (Qwen Team et al., 2024). Evidence for OWL on them is mixed: with SparseGPT at 70%, it nearly halves Llama-3-8B perplexity, from 85.84 to 48.07 (Sieberling et al., 2025). But OWL slightly hurts a 1B Llama 3 model at 50% sparsity in a multilingual setting (Choenni and Titov, 2025), and the layer-wise allocation of Jamshidi and Shvets (2026) gives only marginal gains on Llama-3-8B. ATP (Huang et al., 2025) even argues for a schedule that simply increases with depth. Systematic tests across sizes within one modern family, especially below 3B, remain limited. We ask whether OWL's benefit still grows as models shrink, and whether it comes from aligning sparsity with outlier-rich blocks.

E1: Scaling within a family. We prune Qwen2.5 0.5B, 1.5B, 3B and 7B, and Llama-3.2-1B, Llama-3.2-3B and Llama-3.1-8B, at 50%, 60% and 70% with Wanda and SparseGPT, each with uniform and OWL allocation, using the paper's 7B setting (M=5, λ=0.08). Small models start from higher perplexity, so besides the absolute drop we report the OWL gain in log perplexity, $\Delta L = \log \mathrm{PPL}_{\mathrm{uni}} - \log \mathrm{PPL}_{\mathrm{OWL}}$, separately per method, plus the loss relative to dense and mean zero-shot accuracy at 70%. Size also changes width and depth, so this is a trend, not a causal effect. All blocks in a model are equal in size, so block sparsities average to the global one, which we also measure.

Controls. To test whether the position of each block's sparsity matters, we randomly permute OWL's block sparsities (three permutations per model, Wanda, 70%), keeping the values but breaking their link to the LOD. Two extra calibration seeds for Qwen2.5-0.5B and Llama-3.2-1B check the sensitivity to calibration samples.

E2: Does the LOD predict the gain? We compute the LOD of all ten dense models (three LLaMA-1/OPT, seven modern). We measure unevenness with the coefficient of variation across blocks and the U shape with U, the mean LOD of the outer 20% of blocks minus that of the middle 60%. With ten related models, Spearman correlations with ΔL (per method, with bootstrap intervals, leaving out one family at a time) are exploratory.

Exp.	Models	Runs	A100 h
R1	LLaMA-7B/13B, OPT-6.7B	15	3
R2	LLaMA-7B	28	2
R3	LLaMA-7B	6	3
R4, E2	10 dense models	10	1
E1	7 Qwen2.5 / Llama-3	113	15
E3	2 small models	40	2
Total		212	26

Table 1: Planned runs (pruning plus evaluation, excluding dense baselines) and estimated A100 hours.

E3: Hyperparameter transfer. A weak result could just mean bad settings. We run the paper's grid ($M \in \{3, 5, 7, 10\}$, $\lambda \in \{0.02, 0.05, 0.08, 0.1, 0.2\}$) on Qwen2.5-1.5B and Llama-3.2-1B at 70% with Wanda, select on the C4 validation shard (disjoint from the calibration shard), apply the choice to the whole family, and report default and tuned OWL.

4 Resources

No paid inference API is needed: all models are open-weight on Hugging Face.

Memory. A 7B model peaks at about 18 GB in fp16 (13.5 GB of weights plus one block's calibration activations), so 7B and 8B models need a 24 GB GPU, LLaMA-13B about 31 GB, and models up to 3B fit in 16 GB.

Compute. From the paper's timings (266 s for SparseGPT on 7B, under 1 s for the Wanda metric), Table 1 estimates 26 A100 hours. We budget 60 for zero-shot evaluation and reruns. We use USF's GAIVI cluster (15 A100, 6 H100, 10 L40S), free with a supervisor's approval.

Colab Pro. Both authors have Google Colab Pro (100 compute units a month each, A100 40 GB and L4 GPUs), worth 13 to 37 A100 hours a month at 5 to 15 units per hour. It is available now and fits every model up to 13B. Kaggle, Modal credit or 60 rented hours (under $110) are backups.

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